CMPU2032 - Programming & Algorithms 2 (2025/26)

Extra Credit Report

Student Name: *Dwight Egerton*

Student ID: *D24127624*

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Overview

This is an extra credit assignment; although it is graded out of 10 marks, the overall course total remains capped at 100, so the maximum achievable overall mark will still be 100.

- **Task: Create a Python program using SQLite3**
    - Perform basic CRUD operations for a employee management system.
    - Buold an interfact to demonstrates these operations.

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Setup and Execution

Create Virtual Environment

To ensure this project does not impact any other parts of your system, a virtual environment should be setup. The assumption is that you have Python (version 3.11 or later) installed on your computer. This can simple be done by executing the following commands: -

- Windows ...

```
python -m venv .venv
.venv\Scripts\activate
```

- Mac, Linux or WSL ...

```
python3 -m venv .venv
source .venv/bin/activate
```

This can be done in other ways, for example: through your IDE or with tools like `pipx`. Please refer to the respective documentation for any other approach you wish to follow.

Install dependencies

This project uses `Poetry` for dependency management, building, running and testing. Run the commands below to install `Poetry` into your Python virtual-environment as-well-as all the required project dependencies: -

```
pip3 install poetry
poetry install
```

Launching the `Smart Hospital System` Application

To run the command-line interface: -

```
poerty run main
```

Executing all Application Code Unit-Tests

To run the unit tests: -

```
poetry run tests
```